LABORATORY REPORT: DIVISION OF FORENSIC BIOLOGY

Case Reference: HUM-TH01-2026
Date of Analysis: October 7, 2026
Examiner: Ummay Habiba , Forensic Biology Specialist (In Silico)

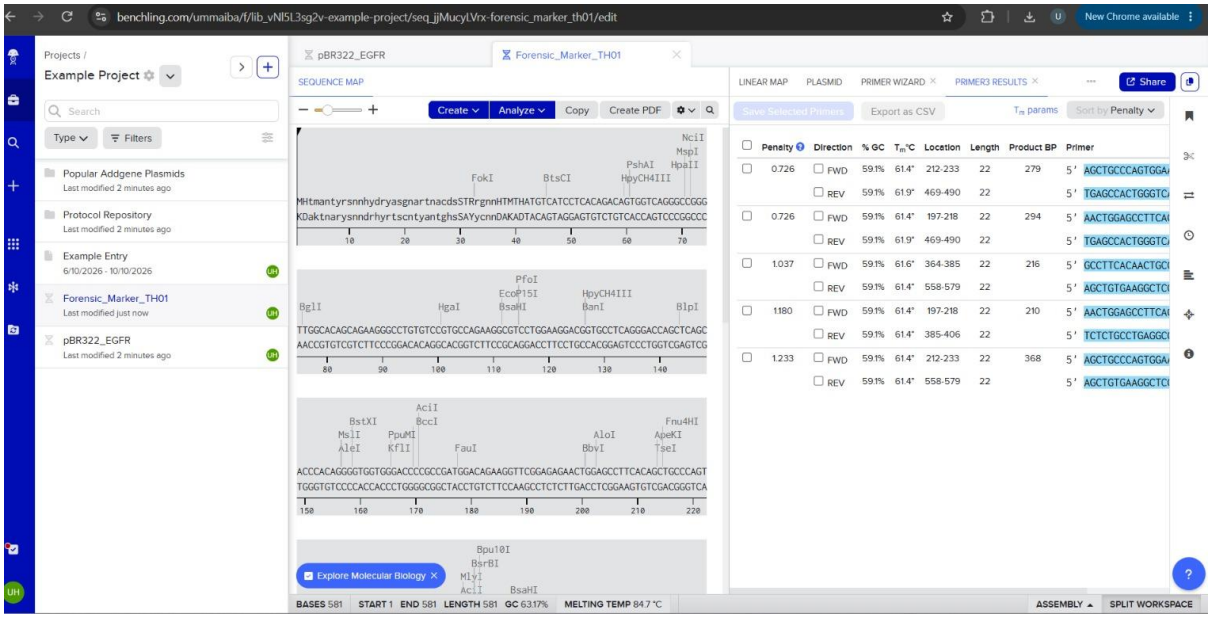
1. OBJECTIVE OF EXAMINATION
To computationally design specific, high-efficiency PCR primer pairs to successfully target and amplify the core human short tandem repeat (STR) locus HUMTH01 for downstream genetic identity testing and profiling.

2. MATERIALS AND METHODOLOGY

- **Target DNA Locus:** Human tyrosine hydroxylase gene region reference profile (M25811.1).
- **Design Platform:** Benchling Core Analytics Framework integrating the Primer3 calculation engine.
- **Thermodynamic Threshold Rules:** Primer length 18–25 bp, target \(T_{m}\) range 55–62°C, optimal GC content ratio 40–60%.

3. OBSERVATIONS & COMPUTATIONAL OUTPUT
The Primer3 engine processed the 581 base pair template file and generated highly stable structural candidates. The optimal premier candidate set selected for report entry demonstrates excellent balance parameters:

- **Forward Primer (FWD Sequence):** 5'- AGCTGCCCACTGGACAG-3' (Location: 212–233)
- **Reverse Primer (REV Sequence):** 5'- TGAGCCCACTGGTCAGC-3' (Location: 469–490)
- **Melting Temperature (\(T_{m}\)):** FWD: 61.4°C | REV: 61.9°C (\(\Delta T_m\): 0.5°C)
- **GC Content Ratio:** 59.1%
- **Expected Diagnostic Amplicon Length:** 279 base pairs (bp)



4. FORENSIC CONCLUSION

The custom-designed primer pair successfully meets all global diagnostic criteria. The minimal [delta Tm]variance (0.5°C) prevents structural primer-dimer interference during thermal cycling, while the 59.1% GC composition ensuring a stable 3' GC clamp guarantees robust target specificity. This virtual configuration is ready for physical assay deployment to isolate individual human STR lengths for profile comparisons.

Signature of Examiner:

[Ummay Habiba]